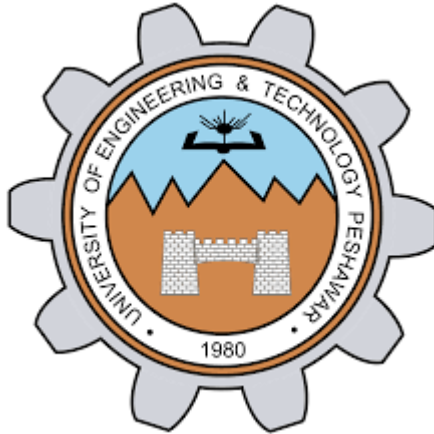




Lab Report – EE-170L Computer Programming (Lab 13)

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Objectives

- To understand the concept of structures in C++.
- To learn how to declare and initialize structures.
- To practice accessing structure members using dot notation.
- To apply structures for storing and displaying real-world data.

Theory

- A structure is a collection of variables (members) of different data types grouped under one name.

- **Syntax:**

```
struct StructName {  
    int id;  
    float value;  
    char name[20];  
};
```

- Unlike arrays (same type only), structures can hold mixed types.
- Structures can be nested, copied, and passed to functions.
- Members are accessed using the dot operator (.).

Lab Tasks

Task 1: Structure Declaration and Initialization

```
#include <iostream>    // tell the compiler to include input output stream library

using namespace std;    // tells the compiler to use standard namespace


struct Person {        // structure declaration

    string name;        // member: name

    int age;            // member: age

    string address;     // member: address

};


int main() {           // main function starting point

    Person person1 = {"Naeem ul Haq", 19, "Peshawar"};    // initialize structure

    cout << "Name: " << person1.name << endl;            // display name

    cout << "Age: " << person1.age << endl;                // display age

    cout << "Address: " << person1.address << endl;        // display address

    return 0;          // tells the compiler that progrm has ended successfully

}
```

Task 2: Student Structure

```
#include <iostream>    // tell the compiler to include input output stream library

using namespace std;   // tells the compiler to use standard namespace


struct Student {       // structure declaration

    string name;        // member: name

    int age;            // member: age

    char grade;         // member: grade

};


int main() {           // main function starting point

    Student student1 = {"Ali Khan", 20, 'A'};    // initialize student

    cout << "Name: " << student1.name << endl;  // output message

    cout << "Age: " << student1.age << endl;    // output message

    cout << "Grade: " << student1.grade << endl; // output message

    return 0;          // tells the compiler that program has ended successfully

}
```

Task 3: Accessing Structure Elements

```
#include <iostream>    // tell the compiler to include input output stream library

using namespace std;   // tells the compiler to use standard namespace
```

```
struct Book {                                // structure declaration

    string title;                            // member: title

    string author;                          // member: author

    float price;                            // member: price

    int pages;                             // member: pages

};

int main() {                                // main dunction starting point

    Book book1 = {"C++ Programming", "Bjarne Stroustrup", 999.99, 1200}; //struc1

    Book book2 = {"Data Structures", "Mark Weiss", 750.50, 850}; //struc2


    cout << "Book 1: " << book1.title << ", " << book1.author    //output message
        << ", Price: " << book1.price << ", Pages: " << book1.pages << endl;

    cout << "Book 2: " << book2.title << ", " << book2.author    //output message
        << ", Price: " << book2.price << ", Pages: " << book2.pages << endl;


    return 0;                               // tells the compiler that program has ended successfully

}
```

Results

- Successfully declared and initialized structures with multiple members.
- Demonstrated accessing structure members using dot notation.
- Showed how structures can store different types of data together.
- Applied structures to real-world examples (Person, Student, Book).